

# Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

**Output Form definition:** Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

MILU evaluates models using accuracy on closed MCQ items via log-likelihood (non-API) or structured JSON parsing (API) [Q48, Q49, Q51, Q52, Q53]. The deployment requires evaluating open-ended Bengali story generation, which demands free-form text output assessed for narrative coherence, register appropriateness, cultural authenticity, and grounding in Bangladeshi named entities. No MILU mechanism produces a score that transfers to story generation evaluation [DATASET-D5 concern]. The authors themselves note log-likelihood may diverge from generation-based and CoT methods [Q85]. The output-form gap is structural: MCQ accuracy measures forced-choice factual recall, not generative cultural grounding. Even if MILU scores were perfectly calibrated to Bangladeshi knowledge (which they are not), they would still not validate story generation quality.

| ID  | Evidence                                                                                                                                                                                                                            |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q48 | "For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)                                                                          |
| Q49 | "We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020)." (p.5)                                                                |
| Q51 | "For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)                                          |
| Q52 | "The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities." (p.5)                                                                                                   |
| Q53 | "We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)                                                                                                |
| Q85 | "Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9) |
| D5  | ৬৫তম ফিল্মফেয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফি জন্য পুরস্কারটিকে জতিছেনে? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry                                                |

## Strengths

- MILU's evaluation infrastructure is well-documented and reproducible across 0/1/5-shot conditions, log-likelihood, and generative-JSON modes [Q46, Q48, Q53], spanning 42–45 models [Q44, Q75]. This breadth means baseline factual-recall comparisons across models are feasible as a screening step.
- Both base and instruct variants are evaluated [Q45], and per-subject/per-language tables are provided [Q99, Q128], enabling fine-grained baseline comparisons within MILU's intended construct.

| ID  | Evidence                                                                                                                                                                         |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q44 | "We evaluate 42 different models on MILU, including large proprietary models, open-source multilingual models, and popular fine-tuned models specific to Indic languages." (p.5) |
| Q45 | "Both the base versions and instruction fine-tuned variants of these models, wherever applicable, are evaluated to measure the improvements gained from fine-tuning." (p.5)      |
| Q46 | "All models, except for proprietary models and LLAMA-3.1-405B, are tested under 0-shot, 1-shot, and 5-shot setups." (p.5)                                                        |
| Q48 | "For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)                       |
| Q53 | "We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)                                             |

|      |                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Q75  | "We evaluate 45 different LLMs and find that the majority of LLMs struggle on MILU, with GPT4o achieving the highest average accuracy." (p.8) |
| Q99  | "Table 9: Detailed subject level statistics of MILU across different languages." (p.20)                                                       |
| Q128 | "Table 47: Detailed subject-wise evaluation for GOOGLE/GEMMA-2-9B-IT on MILU across different languages." (p.52)                              |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                    |
|------|--------------------------------------------------------|
| OF-1 | Does output modality match regional deployment needs?  |
| OF-2 | Assess text-to-speech availability for speech output   |
| OF-3 | Consider literacy rates and accessibility requirements |
| OF-4 | Document form mismatches harming external validity     |

### Checklist responses:

**OF-1:** No — MILU produces single discrete labels (option1–4) or structured JSON [Q51, Q53]; the deployment requires open-ended free-form Bengali narrative text. Output modalities do not match.

**OF-2:** Not applicable — deployment is text-only; no TTS requirement. INSUFFICIENT DOCUMENTATION on whether MILU's evaluation infrastructure could be extended to speech, but this is not a deployment requirement.

**OF-3:** Target population is highly literate (university students [WEB-3]); literacy is not a barrier for either MILU's or the deployment's text format.

**OF-4:** Severe external-validity violation: MCQ accuracy on India-centric content does not predict Bangladeshi story-generation quality. The deployment must construct a separate generative evaluation rubric (per BanglaSocialBench precedent [WEB-16]).

| ID     | Evidence                                                                                                                                                                                                                            |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q51    | "For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i   \cdot))$ " (p.5)                                  |
| Q53    | "We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)                                                                                                |
| Q85    | "Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9) |
| WEB-3  | University student literacy supports text-based deployment ( <a href="https://en.wikipedia.org/wiki/Education_in_Bangladesh">https://en.wikipedia.org/wiki/Education_in_Bangladesh</a> )                                            |
| WEB-16 | BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric ( <a href="https://arxiv.org/abs/2603.15949">https://arxiv.org/abs/2603.15949</a> )     |

## Information Gaps

- No story-generation evaluation methodology exists in MILU; supplementation is structural, not incremental

## Requires Expert Verification

- Design and validation of a Bangladeshi story-generation rubric by native-speaker linguists and literary scholars

## Remediation

**Gap 1: MCQ accuracy via log-likelihood [Q51] does not transfer to evaluating open-ended Bengali narrative generation**

**Recommendation:** Adopt a generation-based evaluation protocol (per the limitation noted in [Q85]) with multi-criteria human scoring (cultural authenticity, register, named-entity accuracy, narrative coherence) plus optional LLM-as-judge calibration validated against Bangladeshi annotator scores. Treat MILU accuracy purely as an upstream screening signal.

| ID  | Evidence                                                                                                                                                                                                                            |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q51 | “For multiple choice questions, given k possible answer strings, we select the answer string (a <sub>i</sub> ) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i   \cdot))$ ” (p.5)                     |
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